

# SHIBLI AFAQ

Smart & Sustainable Cities Researcher | Urban Data Scientist | Architect

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## SUMMARY

Licensed Architect with 4+ years of project delivery experience, now focused on urban climate resilience through smart city research. Completing an M.Sc. in Smart and Sustainable Cities at KFUPM (GPA 4.0/4.0), with research applying GIS, satellite remote sensing, and machine learning to map and mitigate Urban Heat Islands. Proficient in ArcGIS Pro, Python, Google Earth Engine, Interactive Dashboards, and Research. Actively seeking roles in smart city analytics, GIS research, or sustainable urban planning.

## EDUCATION

### King Fahd University of Petroleum and Minerals

Master of Science in Smart and Sustainable Cities

Specialisation in Smart City, Urban Informatics & Sustainability

Cumulative GPA: 4.0/4.0

Thesis: Smart Digital Twin Framework for Urban Heat Island Monitoring, Forecasting, and Mitigation

Relevant Courses: Big Data Analytics; Smart City Systems & IoT; GIS & Spatial Analysis; Environmental Economics; Urban Informatics; Circular Economy

### Birla Institute of Technology (BIT), Mesra

Bachelor of Architecture (B.Arch.)

Cumulative GPA: 7.61/10.0 (First Class with Distinction)

Awards: 1st Place Interior Design Trophy (Zonasa 2019); 2nd Place Café Design (NASA 2017); Shortlisted Main Design Trophy (Zonasa 2018); National Top 66: Embark Competition (ANDC 2019)

## PROFESSIONAL EXPERIENCE

### Metarch Studios

Project Architect

Ranchi, India

Mar 2022 – Aug 2025

- Led full project lifecycle for a **\$720K portfolio** across commercial (restaurants), educational (schools), and residential projects; achieved **100% on-time and on-budget** delivery
- Improved client satisfaction by **20%** above baseline through BIM workflows and advanced 3D visualisation
- Managed interdisciplinary teams (architects, engineers, landscape designers, contractors) across 4 simultaneous project types; reduced construction change orders by **8%**
- Boosted team drawing delivery speed by **15%** through design standardisation and workflow optimisation

### Medicfibers

Graphic Designer

New Delhi, India

May 2021 – Apr 2022

- Created strategic pitch decks and brand materials supporting major investment funding campaigns
- Developed social media strategy increasing brand visibility by **40%** and audience engagement by **3×**

### Metarch Studios

Architectural Intern

Ranchi, India

May 2019 – Jun 2019

- Produced CAD working drawings and 3D models/renderings for interior and exterior design projects

### Chadda and Associates

Architectural Intern

Ranchi, India

May 2018 – Jun 2018

- Contributed to floor plans, sections, elevations, and construction drawings for private and government projects

## RESEARCH & PROJECTS

- Smart Digital Twin Framework for Urban Heat Island (UHI) Monitoring** Aug 2025 – Aug 2026  
M.Sc. Thesis, KFUPM | Vision 2030 aligned; GCC-scalable  
Real-time satellite data pipeline (<10-min latency); 48-hr Prophet ML forecasting (<1.5°C RMSE); Interactive dashboard; Heat Vulnerability Index across 5 Saudi cities; identifies 100,000+ at-risk residents; quantifies 2–5°C cooling potential of green infrastructure interventions
- Smart City Big Data Projects: IoT Pipeline & Multi-City UHI Analysis** Aug – Dec 2025  
Course Projects, ICS 574: Big Data Analysis, KFUPM [Live Demo ↑] [Live Demo ↑]  
IoT Pipeline: 5-layer Kafka/Spark streaming architecture; 300+ concurrent sensors; <10-min latency; ML forecasting ( $R^2 = 0.876$ ); Docker deployment  
UHI Analysis: 25,905 Landsat/Sentinel-2 measurements across 3 Saudi climate zones; NDVI–LST correlation  $r = -0.995$  ( $R^2 = 0.990$ ); Kepler.gl 3D priority intervention mapping

- **GIS & Remote Sensing-Based UHI Assessment: Dammam Metropolitan Area** Jan – May 2026  
*Term Paper, CRP 583: Urban Informatics, KFUPM | Co-author: Sultan Aldhafeeri*  
 5-step GIS/remote sensing workflow across 12,954 grid cells (3,238 km<sup>2</sup>); Landsat 8/9 LST (2020–2025); NDBI identified as dominant LST predictor ( $R^2 = 0.511$ ); Composite HVI (92.6% High vulnerability); dual-threshold mapping isolated 57.3 km<sup>2</sup> urgent intervention zone; Composite HVI hotspot 4.18× larger than LST-only analysis
- **ITS-Based Congestion Management: Aramco Stadium Corridor, Al Khobar** Jan – Apr 2026  
*Term Paper, CE 584: Intelligent Transportation Systems, KFUPM | Team of 4*  
 Evaluated 4 ITS alternatives (Adaptive Signal Control through Congestion Pricing) via structured scoring matrix; recommended integrated ATMS/ATIS/Smart Parking framework; projected 20–35% travel time reduction and 15–25% emissions cut; SAR 28–47M implementation plan; benchmarked against Qatar 2022 and London 2012
- **Twin Tower Complex: Mixed-Use Development** Jan 2020 – Jun 2021  
*B.Arch. Thesis, BIT Mesra*  
 Designed 69-storey net-zero mixed-use complex with earthquake-resilient structure, passive energy systems, full BIM model, and municipal documentation package

## PUBLICATIONS

- SOUNDSCAPE AND THERMAL COMFORT IN HOT-ARID CITIES: A SYSTEMATIC REVIEW** 2025  
*Status: Under Review*  
 PRISMA 2020 systematic review; screened 1,011 studies on acoustic–thermal comfort interactions in hot-arid urban spaces; developed the Integrated Thermal-Acoustic-Perceptual (ITAP) framework for evidence-based public space design in extreme-heat climates
- GIS AND REMOTE SENSING-BASED ASSESSMENT OF URBAN HEAT ISLAND IN DAMMAM** 2026  
*Status: Manuscript in Preparation | Co-author: Sultan Aldhafeeri*  
 GIS and satellite-based HVI framework for the Dammam Metropolitan Area; NDBI confirmed as dominant LST driver ( $R^2 = 0.511$ ); dual-threshold exposure mapping identifies 57.3 km<sup>2</sup> urgent intervention zone; multi-factor hotspot 4.18× larger than temperature-only analysis
- ITS-BASED CONGESTION MANAGEMENT FOR EVENT CORRIDORS IN ARID CITIES** 2026  
*Status: Manuscript in Preparation | Co-authors: H. Alnasser, H. Alghamdi, S. Aldhafeeri*  
 Integrated ITS strategy for event-day congestion on the Aramco Stadium Corridor; multi-criteria evaluation of 4 ITS alternatives; recommends ATMS/ATIS/Smart Parking framework; projects 20–35% travel time reduction within SAR 28–47M; benchmarked against Qatar 2022 and London 2012

## LEADERSHIP & ACTIVITIES

- Student Society of Architecture, BIT Mesra** Ranchi, India  
*Vice President* 2016 – 2021  
 Orchestrated events for a 300+ member organisation; led Vajra Pavilion 2018 (30+ volunteers, on-budget delivery)
- National Service Scheme, BIT Mesra** Ranchi, India  
*Executive Member* 2016 – 2021  
 Monitored the welfare of 10 villages; organised educational initiatives, blood drives, and medical camps

## CERTIFICATIONS

- Licensed Architect: Council of Architecture, India (CA/2021/136485)** Jun 2021  
**Associate Member:** Indian Institute of Architects (IIA), Jharkhand Chapter

## ADDITIONAL

**Spatial & GIS:** ArcGIS Pro, Google Earth Engine (GEE), Kepler.gl, Remote Sensing (Landsat, MODIS, Sentinel-2), NDVI/NDBI Analysis, Heat Vulnerability Mapping, Getis-Ord Gi\* Hotspot Analysis, Dual-Threshold Exposure Mapping

**Data Science & Programming:** Python, Apache Kafka, Apache Spark, Machine Learning (Prophet, Forecasting), Streamlit, Plotly, Docker, GitHub

**Architecture & Design:** AutoCAD, Revit (BIM), SketchUp, Rhino, Technical Documentation, Photoshop, InDesign

**Research Methods:** Systematic Literature Review (PRISMA 2020), Spatial Statistics, Pearson Correlation & Regression, Design Science Framework, Thematic Content Analysis

**Languages:** English: Professional Working; Hindi: Professional Working; Urdu: Native